In order to attain medication safety in polypharmacy management, a strategy could be developed using the strategic framework of the third WHO Global Patient Safety Challenge: *Medication Without Harm*. The key domains to be addressed as part of this framework include patients and the public, medicines, health care professionals, and systems and practices of medication. The four domains and the cross-cutting theme of monitoring and evaluation are elaborated in the following sections, addressing a health system approach for implementing polypharmacy programmes under the umbrella of the Challenge.

4.1 Patients and the public

The role of patients in delivering appropriate polypharmacy

Raising patient awareness about the problems of polypharmacy and non-adherence is important, as patients can play a key role in the prevention and early detection of inappropriate polypharmacy (*11*, *21*). Patients should be seen as shared decision-makers on the use of medication, and health care professionals need to support patients, families and caregivers in order to enable them to undertake this role (*8*). Patients should be encouraged and supported to disclose all the medications they are taking, including OTC and traditional and complementary medicines, especially if they are suffering from multiple conditions and are being treated with polypharmacy (*94*). Health literacy, social norms and cultural factors would need to be taken into account when considering the role of patients, and designing materials for patient education (*95*, *96*).

Patient tools and materials can support the engagement and empowerment of patients, families and caregivers in playing an active role in health care. Such resource materials may include:

- Patient-held medication list or patient-held medication record, sometimes called medication passport (either paper or electronic), can help to optimize patients' medicines. The use of these tools has received positive feedback from patients (*97*). Such lists, provided that they are up-to-date, can also be helpful at care transitions (*98*).
- Patient resource materials that enable patients to understand how to make decisions regarding management of their health conditions and their medications are available (*99*, *100*). An example of a tailormade tool is the *Medicine Sick Day Rules card*, which explains to patients which medicines should be temporarily stopped during dehydration due to an illness (*101*). Different organizations, including WHO, have developed materials that include simple questions to encourage patients to be active participants in their therapeutic decision-making (*102–104*). For example, the *5 Moments for Medication Safety* focuses on five key moments in the medication use process, where action by the patient, family member or caregiver can greatly reduce the risk of harm associated with the use of medications. The tool aims to engage and empower patients to be involved in their own care in a more proactive way and feel responsible for it, encourage their curiosity about the medications they are taking, empower them to communicate openly with their health professionals and be involved in shared decision-making (*104*).

Technology could improve both patient experience and medication adherence. Furthermore, it has potential to enable patients to be active participants in medication reviews. However, more research is required to evaluate strategies for integrating such tools into clinical practice and to ensure they meet their potential in improving patient outcomes and creating value for all users (*105*, *106*).

Prioritizing patients for medication reviews in polypharmacy

Due to limited resources in most health care systems, patients who may benefit from medication reviews need to be prioritized. There are many factors that could increase the likelihood of predisposing a patient to harm

from medication. Research has shown that patients on multiple medications have a higher risk of medication-related harm, due to harm from occurrences such as drug–drug interactions, falls and adverse events (*8*, *48*).

Any methods for identifying patients at risk due to polypharmacy need to be considered in the clinical context. Even in resource-limited settings with inadequate health information databases, the following criteria can be applied to identify situations where a medication review may prove beneficial:

- residents in a care home setting (*12*, *21*)
- patients on high-risk (high-alert) medications (see Annex 1) (*12*)
- patients taking 10 or more medicines (*12*, *21*, *107*)
- patients with two or more co-morbidities (*12*)
- patients with frailty (*12*, *77*, *20*)
- patients with dementia (*12*, *108*)
- palliative care situations (*8*, *12*)

4.2 Health care professionals

Health care professionals and policy-makers can take a leadership role in raising awareness among peers regarding the role of medication review in reducing the harm associated with inappropriate polypharmacy (*11*).

Development of multidisciplinary workforce through changes in undergraduate, postgraduate, and continuing professional development has been identified as a key area to address polypharmacy (*22*). Educational curricula for health care professionals, including physicians, pharmacists and nurses, should include safe medication management to develop necessary competencies and skills for addressing risks associated with polypharmacy (*109*). To help support education of health care professionals, a set of clinical case studies are presented in Annex 4.

Existing guidelines that incorporate key steps for patient-centered medication review should be shared to ensure the dissemination of best practices, including the importance of sharing the information about the outcomes of the medication review in polypharmacy with relevant health care professionals. Establishing support networks to enable health care professionals in sharing scientific and practical information on polypharmacy may assist in that regard. There are several special interest groups focusing on appropriate polypharmacy available internationally (*110–112*).

Health care professionals working in multidisciplinary teams deliver optimum outcomes for patients. They need to be aware of human factors that may affect their performance and the performance of others (*113*). Knowledge of human factors can be further useful in communication with patients and in shared decision-making in polypharmacy (*109*).

A study from Hawaii revealed that many older persons were uncertain about the information about their medications and did not exhibit confidence on understanding the ADRs associated with use of multiple medications (*114*). This emphasizes the need for interventions that are aimed at enhancing the provider–patient interaction about polypharmacy and medication reviews. Safety culture should be addressed to enable health care professionals and patients to discuss issues of polypharmacy, so that patients feel that it is safe and acceptable to ask questions (*95*). For example, *Choosing Wisely* campaign focuses on promoting dialogue between health care professionals and patients; with the aim to reduce utilization of inappropriate tests and medical treatment, empower patients to ask questions, and help patients to choose evidence-based care (*115*).

4.3 Medicines

Polypharmacy and drug interactions

Patients with polypharmacy might be subject to drug–drug, drug–food, drug–disease or herb–drug interactions (*116*). Detection of drug interactions should be included